Task 1: Guided Activity, Restaurant bill calculator

main.swift Run ▶

```swift
import Foundation
let itemPrice = 450.0
let quantity = 3
let subtotal = itemPrice * Double(quantity)
let hasCoupon = true
let discount = (hasCoupon && subtotal > 1000) ? subtotal * 0.10 : 0.0
let discountedAmount = subtotal - discount
let gst = discountedAmount * 0.18
let total = discountedAmount + gst
let share = total / 4.0
let amount = 1234
let hundreds = amount / 100
let remainder = amount % 100
print("Subtotal: ₹\(subtotal)")
print("Discount: ₹\(discount)")
print("GST: ₹\(gst)")
print("Total: ₹\(total)")
print("Each person's share: ₹\(share)")
print("Hundreds: \(hundreds)")
print("Remainder: ₹\(remainder)")
```

Output

```
Subtotal: ₹1350.0
Discount: ₹135.0
GST: ₹218.7
Total: ₹1433.7
Each person's share: ₹358.425
Hundreds: 12
Remainder: ₹34
```

Task 2: Practice Task, Leap year and range checks

main.swift

Run ▶

```swift
import Foundation
let year1 = 1900
let year2 = 2000
let year3 = 2024
let year4 = 2026
let leap1 = (year1 % 4 == 0) && (year1 % 100 != 0 || year1 % 400 == 0)
let leap2 = (year2 % 4 == 0) && (year2 % 100 != 0 || year2 % 400 == 0)
let leap3 = (year3 % 4 == 0) && (year3 % 100 != 0 || year3 % 400 == 0)
let leap4 = (year4 % 4 == 0) && (year4 % 100 != 0 || year4 % 400 == 0)
print("1900 is a leap year: \(leap1)")
print("2000 is a leap year: \(leap2)")
print("2024 is a leap year: \(leap3)")
print("2026 is a leap year: \(leap4)")
let number = 75
let isInRange = (1...100).contains(number)
print("Number is in range: \(isInRange)")
let n = 24
let isValid = (n % 2 == 0) && (n % 10 != 0)
print("Even and not a multiple of 10: \(isValid)")
let nickname: String? = nil
let displayName = nickname ?? "Guest"
print("Nickname: \(displayName)")
```

Output

```
1900 is a leap year: false
2000 is a leap year: true
2024 is a leap year: true
2026 is a leap year: false
Number is in range: true
Even and not a multiple of 10: true
Nickname: Guest
```

Task 3: Predict Task, What does it print?

```
print(7 / 2)
print(7 % 2)
print(-7 % 3)
print(7.0 / 2)
print(10 - 2 * 3 + 4 / 2)
print(5 > 3 && 2 > 4 || true)
print(1 << 3)
print(6 & 3, 6 | 3, 6 ^ 3)

let x: UInt8 = 255
print(x &+ 1)
```

Output

```
3
1
-1
3.5
6
true
8
2 7 5
0
```